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Guidelines for the Responsible Use of Artificial Intelligence in Research

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1. Introduction

1.1 Background

Artificial Intelligence (AI) includes a diverse array of technologies that can perform tasks or behave in ways that are regarded as 'intelligent.' The Organisation for Economic Co-operation and Development (OECD) defines AI as a 'machine-based system that can make predictions, recommendations, or decisions based on human-defined objectives, with varying degrees of autonomy and impact on real or virtual environments.'

In recent years, significant advancements have been made in a new field known as "generative AI" (GenAI). This technology can create content such as text, images, animations, sounds, and source code from scratch. The process of content generation begins with "prompts," which are simple textual instructions provided by the user in natural language. A well-known and debated example of generative AI is ChatGPT, developed by OpenAI.

The best ways to use AI tools for research are still being determined. However, recent advancements in Generative AI (GenAI) are undeniably creating new opportunities to support and partially automate research processes. It is expected that the variety of AI tools available for research will continue to grow in the future and become increasingly integrated into widely used software.

Given that researchers often have limited resources and time, one potential benefit of using Generative AI (GenAI) is increased productivity, particularly for activities that are time-consuming, repetitive, or require minimal intellectual input. Additionally, GenAI could enhance the quality of scientific work by assisting with bibliographic searches and improving the overall quality of the texts produced. AI tools raise significant ethical questions regarding their responsible use. For example, they can generate documents that are syntactically correct but contain false or misleading information, inaccurate bibliographic references, fabricated data, or errors and biases in data analysis that can be extremely difficult, if not impossible, to identify. Additionally, some of these tools have faced criticism from civil rights organisations for their development practices, such as the unauthorised use of copyrighted material or the exploitation of underpaid workers in vulnerable financial situations. Organisations like the United Nations have highlighted significant environmental issues associated with their operations, such as energy and water consumption.

These guidelines aim to establish a strategic vision for the use of AI tools, offering a clearer understanding of what Ca' Foscari University of Venice considers appropriate in research contexts. The university is dedicated to promoting and supporting the responsible use of AI, particularly generative AI, in research. This includes monitoring the application of AI tools within the institution and disseminating ethical principles, best practices, and necessary safeguards to ensure security, privacy, and copyright protection.

1.2 The Focus of These Guidelines

These guidelines pertain to the use of **generative AI tools for research purposes**, primarily focusing on those that are commercially or publicly available. This includes a wide range of applications, whether broad or specific, as well as various types of input and output, such as text, images, or data. Additionally, these guidelines apply to AI tools that are accessible through specific programs via plug-ins, both online and offline.
The guidelines do not apply to the development of AI tools that are an integral part of scientific activities, nor to their application for administrative purposes. Additionally, AI tools used for data mining are not formally included in the research methods or designs discussed in this document. Lastly, these guidelines do not pertain to teaching activities; for those, please consult the relevant University guidelines.

2. General Principles

The University **encourages the use of AI tools** for various research purposes and has minimal specific restrictions. However, it is important to consider current national and international legislation, especially the European Union's AI Act. This act prohibits the use of AI that poses a threat to human beings (referred to as AI with unacceptable risks) and establishes limits on so-called "high-risk" AI applications, such as those used in medical contexts and biometric monitoring. The General Data Protection Regulation (GDPR) includes provisions, specifically in Recital 71, that establish a right to explanation in areas that significantly affect the rights of data subjects. This right must be taken into consideration. Additionally, it is essential to comply with applicable laws and regulations regarding intellectual property, safety, health, welfare, and the protection of human rights. This includes adherence to the University's Code of Ethics, particularly Article 8, which addresses the ethical aspects of research.

The use of AI tools for research purposes **should always follow these general principles**, which align with the key values of trustworthiness, honesty, respect, and accountability outlined in the European Commission's Guidelines on the Responsible Use of Generative AI in Research:

- **The results generated by AI tools must always be verified** through various control steps. Generative AI systems do not guarantee accuracy and lack mechanisms to ensure the truthfulness of their claims. They may produce what is known as "bias" – grammatically correct **content that appears plausible at first glance but contains false information**. It is essential to seek out original sources and properly attribute them to their creators. As AI tools operate more "autonomously," the associated risks increase. Consequently, as the potential for damage grows, the level of verification required by researchers must also intensify.
- **Individuals using AI are fully responsible for the outcomes and the appropriateness of the research processes involving AI**. Responsibility cannot be transferred to AI tools, and it is essential to stay informed about how these tools are used. This means that everyone employing these systems for their research is accountable for any misuse, such as failing to adhere to the principles of scientific integrity on issues such as plagiarism and proper citation of sources. This principle aligns with the Committee of Publication Ethics (COPE) stance, which asserts that authors are entirely responsible for the content of their work,

including any sections generated by AI tools. Therefore, they can be held accountable for any violations of publication ethics. Additionally, ALLEA's 2023 European Code of Conduct for Research Integrity mandates that the use of AI tools be reported in accordance with the standards relevant to the specific scientific field.

- **Creative ideas and human intellectual contributions are essential**. Researchers can generate their own ideas or gain new insights using AI tools, even if the tools themselves do not directly provide those insights. In terms of ideation, AI tools should be seen as versatile resources that enhance human activities, allowing for the generation of more innovative ideas.
- **Researchers should ensure maximum transparency regarding their use of AI tools in scientific activities**. This means that the way AI tools are used should be described in detail in ad hoc sections of the article (e.g., 'Materials and methods' and/or 'Further information'), specifying the prompts (where relevant), which tools were used and which version. This practice is essential for ensuring the repeatability and reproducibility of research. Additionally, AI tools must be properly cited in scientific publications.
- It is essential to adhere to the **specific rules of your discipline regarding the responsible use of AI tools** and to the **guidelines set by academic journals**. Some journals prohibit the use of AI tools in written texts, while others permit their use under certain conditions. Anyone involved in the writing of a scientific article or any other research output should familiarise themselves with these guidelines before submission and ensure compliance.
- Anyone engaged in research should recognise that **using AI tools can have drawbacks in certain situations**. For example, AI tools might produce text that is generic, vague, ambiguous, or redundant. They can also generate incorrect images or diagrams, fabricate quotes, or omit essential information, among other issues. The lack of specificity and accuracy in the text, as well as the use of AI tools themselves – about which one must always be transparent – can have a negative impact during reviews. Additionally, these systems may exhibit bias or discriminatory behavior because they are trained primarily on content available online. As a result, they may not have access to literature that is printed or written

in less widely spoken languages, or that is not openly accessible or shared since the last AI training.

- Always keep in mind that **any input or prompt you provide may be stored by the AI tool for training purposes and could also be presented to third parties as output**. Therefore, **we advise against entering the following types of information into AI tools**:
  - Personal data that may lead to the identification of individuals, either directly or indirectly.
  - Data that is crucial for future research exploitation or that is (or may be) protected by intellectual property law.
  - Data whose release could raise ethical concerns, such as the risk of inappropriate use or harm to specific groups, including stigmatisation.
  - Data subject to restrictions imposed by existing contracts, such as non-disclosure agreements, or by rules set by research funding organisations.
  - Data covered by agreements with third parties, such as companies.
  - Data protected by copyright or database rights, unless you have explicit permission from the owner. In this respect, please remember that the works of others, even if not published, are covered by copyright.

  As a general rule, we recommend that you review the privacy settings of the AI tools you use and make any necessary changes. It is also important to assess the level of control you have over the service provider in the event of problems or disputes. Take caution if it seems difficult to exercise that control, especially if you need to enforce your rights in another country.
- **Researchers should not use AI tools to write substantive aspects of peer reviews or evaluations of research proposals**, especially when there is a confidentiality constraint. On the other hand:
  - lead authors can use AI tools to generate feedback;
  - the research team can use AI tools to improve style in peer reviews, provided they implement measures to ensure confidentiality.

3. Purposes of AI in research

3.1 Research purposes

In this section, you will find information on how AI tools can be used in research. Specifically, the table below outlines the main purposes for using these tools, with each purpose illustrated by a brief case study. Additionally, we will discuss what you can and cannot do when employing AI tools. Finally, we provide some non-exhaustive examples of tools associated with each purpose.

AI tools can serve specific purposes or multiple functions. However, their performance can vary significantly. It is advisable to use the most up-to-date and high-performance tools, which often requires opting for paid versions through individual or group licenses.

Aim	Types of activity	Case studies	DOs	DON’Ts	Examples of AI tools
To find new literature	Information retrieval	When conducting a study of the available literature, the team also uses an AI tool to identify relevant literature in lesser-known fields (e.g. interdisciplinary research).	- Mention the AI tool in the methodology section of the research. - Use AI freely if you are not in the context of a formal study.	- In your review of sources, do not use articles suggested by the IA without checking their content.	- Elicit (http://elicit.org) - Perplexity (https://www.perplexity.ai) - Inciteful (http://inciteful.xyz) - Research Rabbit (http://researchrabbit.ai) - Connected Papers (http://connectedpapers.com) - Consensus (http://consensus.app) - Scite (http://scite.ai) - Iris (http://iris.ai)
To summarise articles and published papers	Word processing	The research team uses an AI tool to select and summarise recent literature. The summaries are then examined in detail.	- Using AI for initial screening before reading the full texts.	- Do not quote or reference articles based on a summary, without having read them personally.	- Semantic Scholar (http://semanticscholar.org) - Humata (https://www.humata.ai) - Iris (http://iris.ai) - ChatGPT (http://chat.openai.com) - Perplexity (https://www.perplexity.ai) - Copilot (https://copilot.microsoft.com)

Aim	**Types of activity**	**Case studies**	**DOs**	**DON'Ts**	**Examples of AI tools**
Selective searches of an article's content.	Information retrieval	The research team uses an AI tool to identify recommendation s/perspectives in scientific publications to keep up-to-date with the latest implications in their field.	- Get a screening for specific parts of articles or terminology and then do a manual review.	- Do not summarise specific parts of articles without reading them carefully in their entirety. - Do not forget to cite sources you have used in the preparation of your academic work.	- Iris (http://iris.ai) - Semantic Scholar (http://semanticscholar.org) - ChatGPT (http://chat.openai.com) - Copilot (https://copilot.microsoft.com)
To formulate scientific statements / hypotheses	Word processing	The team enters their scientific statements/hypot heses into an AI tool to receive feedback. The tool suggests possible (substantial or clarifying) improvements.	- Mention the use of the AI tool (including prompts and outputs) in the processing of scientific hypotheses, if hypothesis testing is part of the publication.	- Do not take AI's scientific statements and hypotheses as true without proper verification.	- ChatGPT (http://chat.openai.com) - Elicit (https://elicit.org/?via=topaitools) - Copilot (https://copilot.microsoft.com)

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Aim	**Types of activity**	**Case studies**	**DOs**	**DON'Ts**	**Examples of AI tools**
Critique statements or stances	Ideation	The research team uses an AI tool to critically examine a stance by gathering more information on possible counter-arguments.	- Check whether all arguments that undermine your position can be refuted. - Educate students through debate.	- Avoid citing very personal stories to support your position.	- DebateDevil (https://www.debate-devil.com/en) - Elicit (https://elicit.org/?via=topaitools) - Scite (https://scite.ai/?via=topaitools) - ChatGPT (http://chat.openai.com) - Copilot (https://copilot.microsoft.com)

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Aim	**Types of activity**	**Case studies**	**DOs**	**DON'Ts**	**Examples of AI tools**
To carry out a preliminary screening of evidence for/against given scientific claims	Information retrieval	The research team uses an AI tool to screen the results of scientific publications (in terms of positive or negative relationships between certain variables), to deepen the burden of proof for a given hypothesis.	- Do an initial screening of the evidence and shortcomings of the studies, and then examine them in depth. - Do additional screening during the systematic review and meta-analysis of studies. - During this practice, take into account 'publication bias towards positive results'.	- do not use the results of the AI tool as a 'burden of proof' for or against a statement in the public debate. - do not rely entirely on an AI tool when selecting studies for a systematic review.	- Consensus (https://consensus.app/search) - Perplexity (https://www.perplexity.ai) - System Beta (https://www.system.com/landing) - Scite (https://scite.ai/?via=topaitools) - ChatGPT (https://chat.openai.com) - Copilot (https://copilot.microsoft.com)

Aim	Types of activity	Case studies	DOs	DON'Ts	Examples of AI tools
Brainstorming	Ideation	The research team submits its ideas to an AI tool to find out whether they are already covered in the existing literature. The tool finds some potentially useful publications to answer the research query.	- Cite the interactions you had with the AI tool in order to generate ideas for your publication (including prompts and outputs).	- Do not introduce completely new ideas expecting to commit later in their valorisation.	- ChatGPT (http://chat.openai.com) - Elicit (https://elicit.org/?via=topaitools) - Copilot (https://copilot.microsoft.com)
To generate the initial structure and lay out the sections of a scientific publication	Word processing	The research team uses an AI tool to lay out the initial structure for the introductory section of a publication. The team then writes the text based on that structure.	- Cite AI tools among the additional data in the methodological section of the research.	- Do not blindly follow the proposed structure without checking and comparing it with your field's accepted practices.	- ChatGPT (https://chat.openai.com) - Copilot (https://copilot.microsoft.com)

Aim	**Types of activity**	**Case studies**	**DOs**	**DON’Ts**	**Examples of AI tools**
To generate a text by entering the initial structure of a scientific publication	Word processing / new work production	The research team provides paragraph titles with instructions to write N lines per paragraph. The AI tool generates the text accordingly, providing information about the sources.	- Read all cited publications and check whether the citation is accurate. - Check whether the paragraphs present sufficiently diverse views/evidence (e.g. studies with conflicting results). - Report the use of AI tools in the methodology section of the publication and cite prompts and outputs among the supplementary data.	- Do not copy and paste texts into your publications without checking for plagiarism or licencing issues.	- ChatGPT (https://chat.openai.com) - Copilot (https://copilot.microsoft.com)

Aim	Types of activity	Case studies	DOs	DON'Ts	Examples of AI tools
To correct spelling mistakes, grammatical errors and paragraph structure according to your input	Word processing	- The research team uses an AI tool to improve their text. The tool makes suggestions to improve paragraph structure and corrects grammatical and spelling errors. It also suggests alternatives for inaccurate terms.	- Cite AI tools in the methodology section of the publication.	- Do not automatically accept all possible changes without verifying whether your content has changed.	- Writefull (https://www.writefull.com) - Quillbot (https://quillbot.com) - ChatGPT (https://chat.openai.com) - Copilot (https://copilot.microsoft.com)

Aim	**Types of activity**	**Case studies**	**DOs**	**DON'Ts**	**Examples of AI tools**
Analysis of research data	Data analysis and visualisation	Various AI tools are already available for analysing research data. This is not necessarily generative AI; analysis can be carried out based on prompts with existing software.	- Carry out analyses using AI with sufficient substantive control and in accordance with ethical and legal principles (e.g. check bias in data sets). - Cite AI tools in the methodology section of the research.	- do not perform analyses based solely on prompts without checking whether they have been interpreted correctly. - do not make post hoc hypotheses on a dataset because these hypotheses are always confirmed. - GenAI was not designed for this type of application and should be avoided	- Code Interpreter in GPT-4 (https://chat.openai.com/?model=gpt-4-code-interpreter) - AutoML

4. Benefits of AI in Research

AI tools offer many benefits for research, including:

- **Increased productivity**, i.e. the amount of research outputs produced with a given amount of resources. Tasks that are more time-consuming or repetitive can, at least partially, be delegated to AI tools. Moreover, such tools can also strengthen the researchers’ skills in generating ideas or can be used to identify complex patterns within large datasets, which can be particularly useful in exploratory research.
- **Increased quality of work**. The use of GenAI tools can help researchers achieve their objectives more efficiently. For instance, studying scientific literature on a specific topic can be enhanced with the support of artificial intelligence. Additionally, AI tools can inspire insights, creativity, and self-critique, ultimately leading to higher-quality work.
- **Help in assessing one's own work**. AI tools can provide valuable insights into an individual's beliefs, arguments, ideas, and outputs and offer a rapid external and objective perspective.
- **AI can help us face new challenges more readily**. Today, researchers frequently encounter obstacles that demand new skills, such as writing project applications, drafting Data Management Plans, and navigating data protection legislation and ethical guidelines. One major hurdle is that the learning process can be time-consuming. However, utilizing AI tools could help streamline and expedite this process.
- **Bring about a change in our daily work**. Artificial intelligence and other technologies are transforming work roles across various fields. These changes can impact various research activities, from project coordination and mentoring to administrative tasks and basic research. AI tools are anticipated to primarily be used for repetitive, non-intellectual tasks, such as bureaucratic duties. In contrast, AI is often employed alongside humans for more complex tasks. If we assume that AI will reduce the time spent on repetitive tasks without increasing their

volume, it is likely that more complex tasks – such as those involved in basic research – will occupy a greater portion of researchers' daily work in the future.

- **Reduce gaps between researchers with diverse characteristics and backgrounds**. Research is a usually collective effort involving individuals with diverse characteristics and backgrounds. This diversity necessitates a consideration of the different opportunities and challenges faced by vulnerable or disadvantaged groups, such as individuals with Specific Learning Disabilities or other disabilities, neurodivergent individuals, and non-native speakers. For some, the use of AI tools can provide significant benefits. For instance, a researcher who struggles with writing research findings in English may utilise AI tools to help bridge this gap. In other words, AI tools have the potential to level the playing field for researchers by addressing specific skill deficits.
- **Making science more accessible**. AI tools can be used to summarise and explain scientific texts, making them more accessible to a non-academic audience. They also help scientists communicate their results and findings more clearly. These tools effectively address a significant cognitive limitation known as the “curse of knowledge.” This occurs when a specialised individual assumes that others have the same background and understanding as they do, leading to misunderstandings in communication.
- **Promoting dialogue and research across disciplines**. AI tools help researchers explore and understand large volumes of information found in extensive literature across various fields. These tools can also identify connections that may not be immediately apparent to researchers: by acting as matchmakers, AI tools can connect different research groups, thereby promoting interdisciplinary collaboration.

Overall, it is expected that, with equal resources, these benefits for individual research teams will result in **faster scientific progress** and a bigger social impact of research at large. This indicates that the resources allocated to scientific research are likely to be used more effectively.

5. Challenges and Potential Problems in the Use of AI for Research

It is important to recognise that AI tools, especially generative AI, can have significant limitations, whether due to the technical shortcomings of the tools themselves or their potential for misuse. Additionally, risks may stem from the proprietary nature of certain tools, which may involve fees and restricted access to services, or from the concentration of ownership in a small number of companies, leading to unclear or opaque data usage practices. The key critical issues are outlined below by category.

5.1 Outputs of AI Models

- **Veracity**. The results of generative AI, particularly those based on Large Language Models (LLMs), may seem accurate and persuasive, yet they can actually be incorrect, fabricated, biased, or simply irrelevant. In this context, we describe as "**biases**" those instances where an AI tool generates outputs that are either not true, not factually accurate or not based on objective data. Additionally, the outputs produced by AI tools can often be redundant, irrelevant, or overly technical in style. The widespread and uncritical use of artificial intelligence can negatively impact the quality of work, particularly in areas where AI tools are not appropriate. The confidence with which these tools present their findings can be misleading, so it is essential to **always evaluate very carefully** the results they provide.
- **Bias**. AI tools can produce biased and distorted outputs and contain sexist, racist, or other discriminatory content. Developing LLMs for specific tasks that require human evaluation might also introduce bias.
- **Lack of detailed information on how AI tools exactly work**. Determining how algorithms reach specific results can often be challenging. While it may be evident that certain parts of the training data significantly impact outcomes, the reasons behind this influence may not always be clear. A lack of understanding regarding the precise functioning of AI tools can lead to serious consequences, particularly if

unintended correlations arise concerning sensitive characteristics such as gender, sexual orientation, religion, and others, or their indicators.

- **Limited reproducibility**. Due to their stochastic nature, GenAI tools do not always produce identical outputs. For example, if you enter the same prompt multiple times, you may receive different segments of text each time. This unpredictability makes it challenging to achieve complete reproducibility in search processes, such as when you use an AI tool to gather literature on a specific topic. Additionally, algorithms can be unstable; inputs may not be processed consistently over time, leading to difficulties in establishing reproducible workflows.
- **Plagiarised elements within the output**. Copying formal elements or ideas from existing texts or images in their entirety can infringe on copyright and research integrity. Not all AI tools can explicitly reference the sources they use. Citing sources can involve certain complexities, such as when sources appear within other sources (e.g., citations found in a review article) or when a text is misattributed to a specific source.
- **The generated new data is in conflict with existing privacy regulations such as the GDPR.** Issues can occur with data linkage or derived information when individually anonymised data can, when combined, lead to the creation of new privacy-sensitive data (for example, linking income class with postcode). This indicates that the original datasets were not genuinely anonymous, as they were not irreversibly de-identified. Furthermore, contrary to the principle of maintaining accurate and up-to-date personal data, AI systems may produce incorrect or misleading information about specific individuals.

5.2 Training and Learning Data for AI Models

- **Data distortions** may occur in training.
    a. Training data may already reflect **existing structural inequalities** between different groups. Variations among groups, such as those based on gender, ethnicity, or socio-economic class, can lead to either over-representation or under-representation of real-world conditions in the data (for example, in labour force records, health data, etc.). The impact of structural inequalities

on group participation or outcomes is often greatly influenced by the source and context of the data.

b. Training data may contain **discriminatory language, racist overtones, incorrect information, and fallacious arguments** resulting from logical errors. This is especially true when written texts are used as training data for large language models (LLM).
c. Training data may present other forms of bias depending on the source type. For instance, the 'publication plus bias' may impact AI tools that initially screen evidence for subsequent scientific outputs.

- **Training data for AI tools may be dated**. AI tools, including LLMs, may not have access to the most recent events or publications, as their training data has a knowledge cutoff. For example, GPT-3.5's knowledge limit was September 2021, and it did not include sources published after that date.
- Training data for **AI tools may already be partially influenced by large language models (LLMs)**. If the use of LLMs becomes widespread, this could affect the content of new texts, which may then be reused as training data for LLMs. This creates a form of "circularity," posing the risk that, if there is insufficient human intervention, machines may replicate the errors made by other machines.
- **Various segments of training data are not equally informative**. Training data for large language models (LLMs) can include various sources of information that typically have different levels of credibility. For instance, the results of a single empirical study are generally considered less credible than a meta-analysis that compiles all available empirical studies. Similarly, blogs, newspaper articles, and scientific publications do not inherently have the same level of credibility.
  LLMs, like ChatGPT, can theoretically account for these differences. This can occur through refinement phases or reinforcement learning from human feedback (RLHF) during their development. However, many LLMs still have the potential to connect users to less reliable sources, such as blogs, especially if no plug-ins or specialised AI tools are utilised.
- Training data may contain **personal information or copyrighted material** that was unlawfully disclosed. The use of this data could therefore lead to violations of

research integrity, intellectual property rights, and/or the GDPR as well as relevant national data protection laws.

- Training data used by AI tools can sometimes be **reconstructed under certain conditions**. Various types of "adversarial privacy attacks" – which aim to disrupt the functionality of AI models or steal sensitive information – pose significant risks. This is especially true for AI tools that analyse personal data or use images protected by rights without proper release. Under specific circumstances, third parties may be able to request data about particular individuals involved in the training (known as "membership inference attacks"). Additionally, parts of the training data could potentially be reconstructed through "model inversion attacks.".

5.3 Text Prompts and User Input for AI Models

- **Entering personal data into AI tools could potentially breach the European Union's General Data Protection Regulation (GDPR)**. When personal information is inputted, it may be stored outside the EU, and third parties might gain access to this data.
- **Entering personal data into AI tools might conflict with the consent provided by research participants**. Ethical guidelines for research participation often state that consent is deemed informed only if participants are made aware of all risks in advance. If the use of AI tools leads to unforeseen risks that were not disclosed, this may indicate that the consent was not adequately 'informed'. Additionally, informed consent may also imply that participants' information should not be shared with commercial companies.
- **Using AI tools to provide feedback on completely new research ideas may be akin to making those ideas public for specific goals.** For instance, it is important to consider whether employees of an AI tool company can view users' conversations with their tool to enhance the algorithm. If this is the case, it could result in the **loss of opportunities to patent** and commercialise an invention. Therefore, be careful with what you include in discussions if you plan to secure a patent in the future.

- **Your inputs into AI tools can affect the results obtained by others.** Some free AI tools may use content – such as prompts, responses, uploaded images, and generated images – to enhance their services by training on the interactions people have with the tool. This may happen unless you choose to disable this option. Consequently, your interactions with the AI tool, including your prompts, could be used to improve the model's performance.
  It is unclear whether this process involves basic training, refinement, or reinforcement learning from human feedback (RLHF). If it involves basic training, your prompts or inputs might reappear as outputs for other users, although there is currently no conclusive evidence to confirm this. Conversely, if the process only involves further RLHF, your prompts are unlikely to be shared with other users.
- **The data entered into the system, such as quantitative data, may sometimes be of poor quality**. This can result from flaws in the sample itself, which diminishes the data's usefulness for further analysis—this is known as methodological defects. Additionally, the data might lack sufficient context, referred to as metadata, making it inadequate for effective processing by AI models. Moreover, current AI models do not always manage metadata proficiently. Therefore, it is advisable for research teams to **keep in mind the principle of "garbage in, garbage out."**

5.4 Environmental Implications

Certain AI systems, like large language models (LLMs), consume vast amounts of water, energy, and raw materials, leading to significant impacts on both people and the environment. When using these systems for research, it is important to avoid generating unnecessary prompts and to restrict their use to essential purposes only.

- **Water consumption**. The OECD Observatory on AI reports that training a large language model like GPT-3 can use millions of litres of fresh water for system cooling. It is projected that AI could consume up to 6.6 billion cubic meters of water globally by 2027. This is a significant concern, particularly in countries that are experiencing drought.

- **Electricity consumption.** According to the International Energy Agency, a request made through ChatGPT consumes ten times more energy than a search on Google.
- **Consumption of raw materials**. According to the UN report[1], 800 kg of raw materials are required to produce a 2-kg computer. Additionally, the microchips that drive artificial intelligence rely on rare earth elements, which are frequently extracted in environmentally harmful ways.
- **Local environmental damage.** Data centres generate electronic waste that contains hazardous substances and pollutants, including mercury and lead. The International Energy Agency reports that data centres used for AI consume significant amounts of energy, which is often produced by burning fossil fuels. This process releases greenhouse gases that contribute to global warming.
- **Conseguenze per le popolazioni locali**. I data center sono spesso collocati in Paesi in via di sviluppo, che subiscono lo sfruttamento di materie prime, acqua potabile e risorse umane e devono quindi affrontare scarsità di risorse.
  Impact on Local Populations. Data centres are frequently situated in developing countries, which suffer from the exploitation of raw materials, drinking water, and human resources, leading to resource shortages[2].

6. Citing AI Tools in Research

6.1 State of the Art

The rise of AI tools employed in research presents new challenges for attributing scientific work. Currently, there is no consensus on the acceptability of these tools or how to ensure their use is transparent. Academic communities across various disciplines, as well as scientific journals and publishers, hold differing views on the use of AI tools and how they should be cited. It is important to note that **some publishers and journals impose specific restrictions on the use of AI tools, so it is advisable to check these**

[1] United Nations Environment Programme (2024). *Artificial Intelligence (AI) end-to-end: The Environmental Impact of the Full AI Lifecycle Needs to be Comprehensively Assessed - Issue Note.* https://wedocs.unep.org/20.500.11822/46288.
[2] https://inthesetimes.com/article/microsoft-ai-data-center-water-climate.

guidelines before submitting a paper. This inquiry should also be made prior to conducting research if the use of AI tools is a formal part of the research methodology or project. Furthermore, this check should be conducted beforehand if you plan to use AI tools to generate text or images while writing your paper.[3]
Citation practices regarding AI will continue to evolve, so it is recommended to regularly review these aspects.

6.2 Citations and AI

The University recommends that you **check the guidelines of the journal or publisher you plan to submit your work to**, if such guidelines are available.
If the journal or publisher does not provide specific guidelines regarding the use of AI tools, you can follow these general recommendations. It is advisable to describe the use of AI tools in the **methodology section** of your work, particularly when their use is significant and goes beyond simple tasks like grammatical, syntactic, or stylistic corrections. When detailing the use of AI tools, include the following technical specifications: the tool's full name, version, a link to its repository, as well as any configurations and parameters used. If the AI tool has been described in detail in scientific publications, these articles should be cited. The prompts used and the outputs obtained should be described in the methodology section or included as supplementary data. If the AI-generated text contains any fragments or ideas from other sources, it is essential to cite the original sources accordingly.

Please take note of these special instances:

a. **When AI tools are the object of the study (for example, you are analysing distortions in ChatGPT outputs) or you are citing their outputs verbatim**. In this case, it is important to accurately present all prompts and cite the results precisely as they appear. This can be done by including the name of the AI tool in brackets at the end of the quotation. Generally, it is not necessary to list the AI tool as the 'author' in the bibliographical references section, but if the journal's citation style allows it, then it is advisable to do so.

[3] We refer to images that do not describe the research results, because generative AI systems are not designed for data visualisation.

b. **When AI tools generate seemingly new insights or arguments without significant human intellectual contribution.** To avoid plagiarism, it is essential to first verify the origin of any output. If an idea already exists, its source must be properly cited. If a thorough search fails to identify the origin of the idea, it is permissible to present it, but it should be clearly stated that the idea comes from an AI. The methods used to determine the origin should be detailed in the methodology section of the research products, and the prompts should be included as supplementary data.